BIOSOLVE 2026

BinGo

Smart Waste Sorting · SDG 15 — Life on Land

Hackathon: BioSolve 2026

Problem Statement: Smart Waste Classifier App (SDG 15: Life on Land)

Team Name: Fantastic Four

Snap a photo of any waste item — BinGo tells you how to dispose of it, on-device, with no data leaving your phone.

One photo → 5 waste categories

- Recyclable
- Organic
- E-Waste
- Hazardous
- General / Non-recyclable

PROBLEM & SOLUTION

Understanding the Problem

Improper segregation at household level in India sends recyclable and hazardous waste to open landfills.

Environmental damage: heavy-metal leaching, soil degradation, groundwater poisoning, habitat destruction.

Ambiguity at the bin: multi-material packaging, e-waste and household hazardous items are hard to place.

Result: contamination and improper dumping, even when people want to sort correctly.

Our Solution: BinGo

Mobile-first web app: capture or upload one photo of an item.

In-browser AI classifies it into 5 standard waste categories.

Clear India-friendly steps (dry/wet bins, kabadiwalas, e-waste centres).

SDG 15 fact on every result links disposal to soil and biodiversity.

Honest "Not sure" fallback instead of misleading advice.

Project Live Link
BinGo

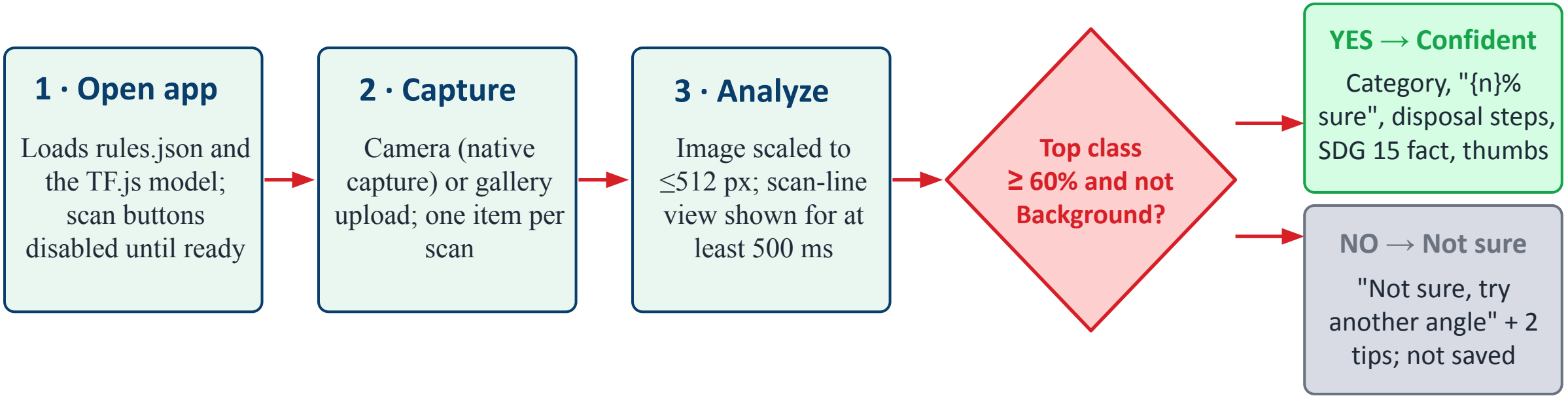
Instant clarity
Result in under 2 s (target)

Zero barrier
No login, no install, no server round-trip

Transparent ML
Confidence %, "Not sure" gate, feedback

Environmental literacy
Disposal tied to SDG 15: Life on Land

HOW BINGO WORKS



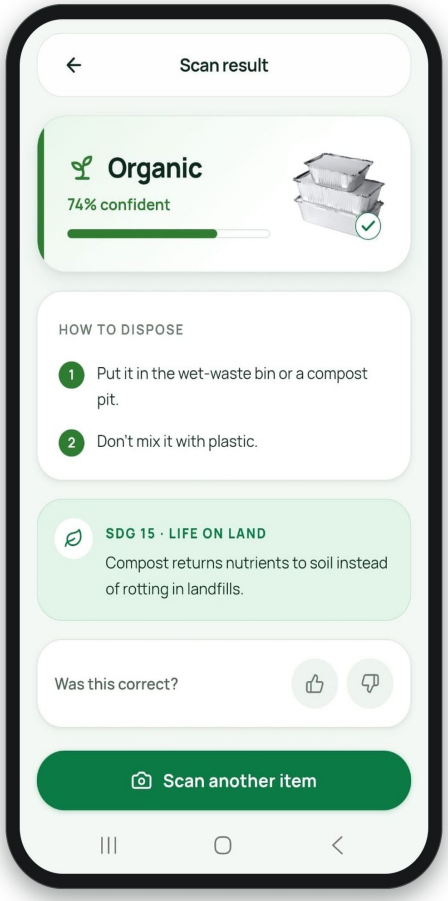
On every confident scan

- **100 px thumbnail** saved to local history (last 50 scans)
- **Items-sorted counter** increments (keeps counting past 50)
- **Daily streak** updated by local calendar date
- **Thumbs up / down** feedback stored on-device only

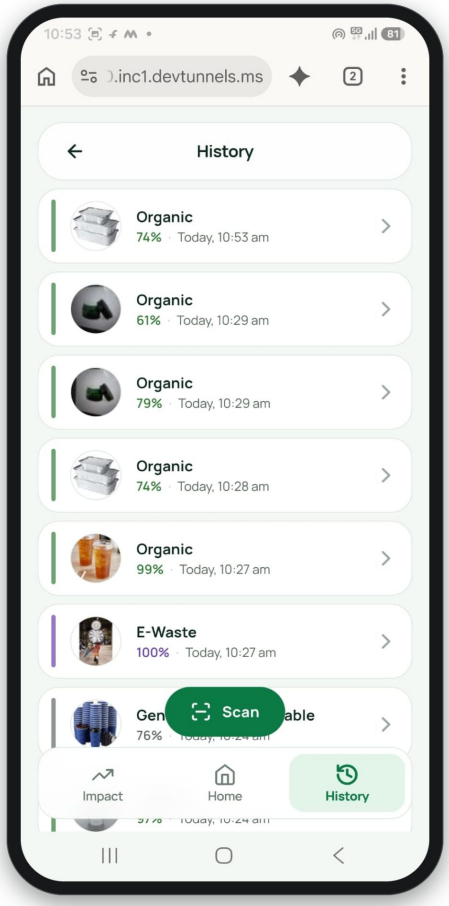
Navigation & next scan

- **"Scan another item"** button on both result screens
- Browser/system **Back** returns to Home or History; never leaves the app
- Close **X** on the analyzing screen returns Home

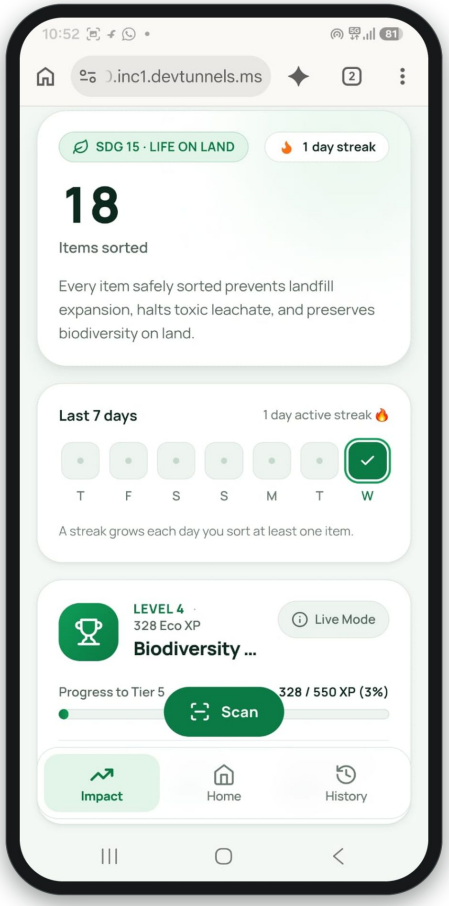
APP SCREENS



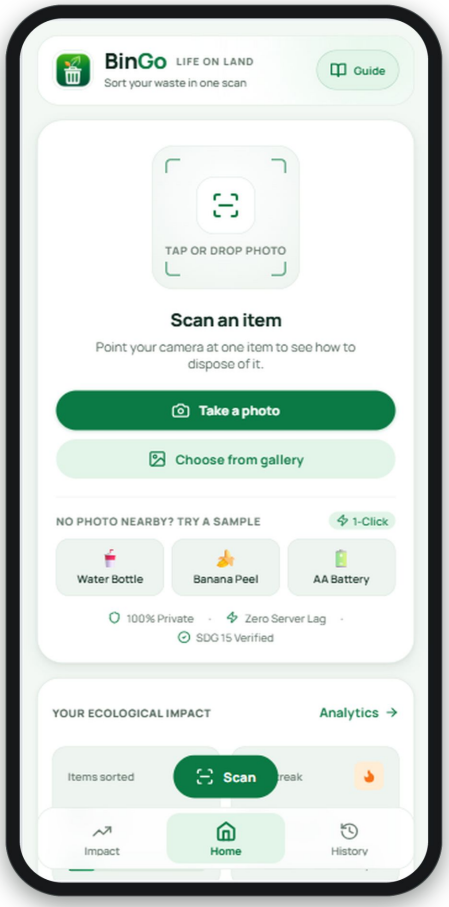
Home — impact card & scan



Result — steps, SDG 15 fact, feedback



Impact Counter



History — last 50 scans

WASTE CATEGORIES & DISPOSAL

Recyclable

Includes

Bottles, clean paper, cans, glass jars, tetra packs

How to dispose

- Empty and rinse it.
- Keep it dry.
- Dry-waste bin, or give to a kabadiwala / recycler.

Organic

Includes

Peels, food leftovers, tea leaves, eggshells, garden leaves

How to dispose

- Wet-waste bin or compost pit.
- Don't mix it with plastic.

E-Waste

Includes

Old phones, chargers, cables, earphones, power banks

How to dispose

- Not in household waste.
- Hand to an authorised e-waste collector or take-back counter.

Hazardous

Includes

Loose batteries, CFLs / tubes, expired medicines, aerosol cans

How to dispose

- Keep separate and sealed.
- Don't burn, flush or mix it.
- Hazardous-waste collection point.

General / Non-recycle

Includes

Chip wrappers, soiled paper, tissues, thermocol, broken ceramics

How to dispose

- Wrap it, general dry non-recyclable bin.
- Don't litter.

Hidden 6th class — Background (empty frames, walls, hands, clutter): never shown to users; if it wins, the app answers "Not sure".

TECHNICAL APPROACH

Technology Stack

Frontend: React 18, Vite 5, Tailwind CSS 3.4, Lucide icons

On-device AI: TensorFlow.js + MobileNetV2 graph model (transfer learning, 224×224 RGB input)

Storage: localStorage with in-memory fallback — streak, total sorted, history[50]

Content: editable rules.json (steps, labels, SDG facts) — no rebuild needed

Hosting: static files on Vercel — **no backend**, no database, no analytics

91.49%	60%	~10 MB	58 kB
test accuracy (target ≥ 80%)	confidence gate for "Not sure"	model weights, served separately	gzipped JS bundle (limit 350 kB)

Github Repo
(Source Code)

Client-side Architecture

React 18 + Vite UI
Home · Scan · History screens

↓

View state machine
home ⟶ analyzing ⟶ result ⟶ history

↓

Decision engine
60% confidence gate + Background filter

↓

TensorFlow.js model
MobileNetV2 · EXIF-aware 224 px input

Runs fully on the phone: no photo or personal data leaves the browser. HTTPS required for camera.

FEASIBILITY & IMPACT

Challenge → Fix

Unknown or unclear items	→	60% gate + hidden Background class return "Not sure" instead of wrong advice
Real phone photos	→	EXIF-aware decoding; run-ID guard cancels stale async scans
Storage limits	→	100 px JPEG thumbnails, 50-scan cap, oldest dropped on quota error
Mid-range phones	→	All on-device; ≤ 2 s target; Android Chrome 111+, iOS Safari 16.4+
Out-of-scope waste	→	Sanitary, bio-medical, textiles, debris, furniture may give "Not sure" or nearest class (accepted v1 limit)

Impact & Benefits

Citizens: one-scan clarity with numbered, India-friendly steps

Habit loop: items-sorted counter, day streak, history, thumbs feedback

Privacy by design: no backend, uploads, accounts or analytics

Land & biodiversity: every result carries an SDG 15 fact

SDG 15 · LIFE ON LAND — "Every scan that lands waste in the right bin is a vote for the planet."